Project 2 Bonus: Fault-Tolerant Gossip Protocols & Percolation Dynamics

Course: COP5612 Distributed Operating System Principles (Fall 2026)
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Bonus Component: Node Crash & Transient Link Failure Models (30% Credit)
Analysis: Percolation Thresholds, Small-World Shortcut Fault-Resilience, and Epidemic Self-Healing

1. Failure Models & Experimental Design

To rigorously assess protocol robustness under adverse network conditions, we implemented two parameterized failure models in project2_bonus.erl:

- **Permanent Node Crash Failures (Parameter: *FailureRate*, P_f *in [0.0, 0.5]*):** A designated fraction P_f of actors suffer crash-stop failures. Crashed nodes permanently halt transmission and discard all incoming messages. We measure the **Coverage Percentage** (proportion of surviving non-failed nodes that successfully receive the rumor) and **Convergence Time** of the surviving subgraph.
- **Transient Connection / Link Drops (Parameter: *DropRate*, P_d *in [0.0, 0.8]*):** All nodes remain alive, but network links are lossy: each transmitted packet is dropped in transit with probability P_d. This evaluates the protocol's self-healing capabilities through randomized epidemic retransmissions.

2. Empirical Observations & Failure Curves

Experiments were conducted on a benchmark network of $N = 196$ nodes (a 14 × 14 lattice for 2D topologies). The figures below plot coverage resilience against node failures, and convergence delay under packet loss.

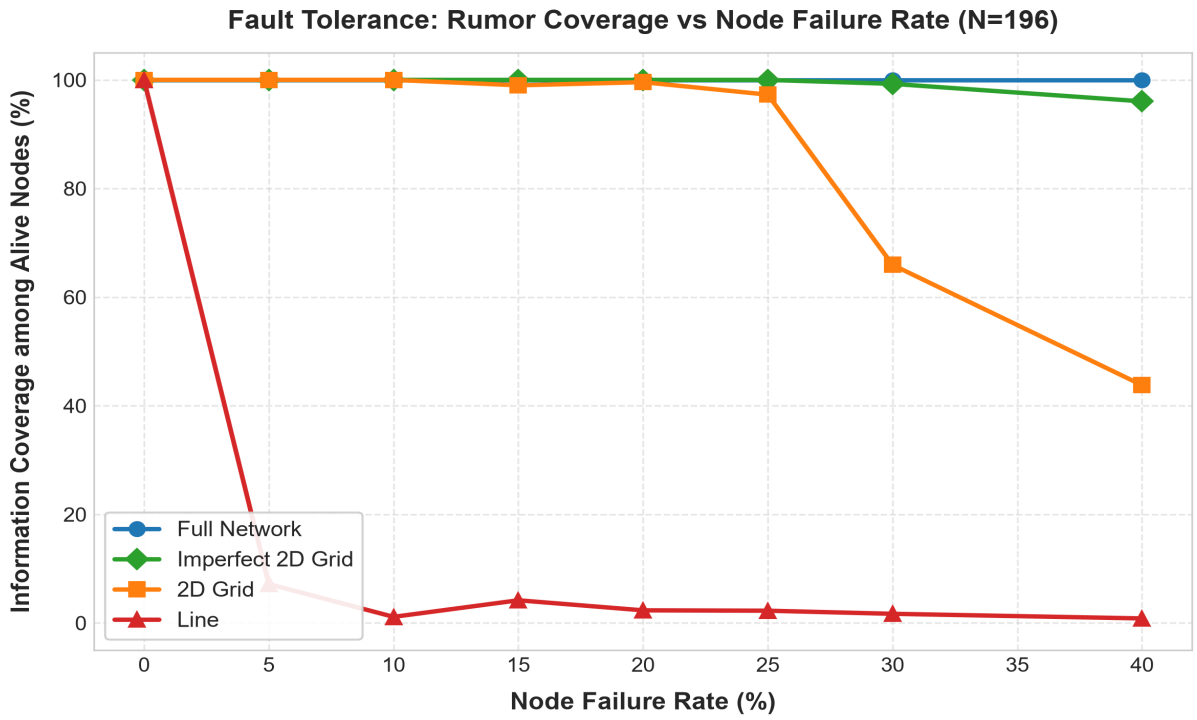


Figure 3: Rumor Coverage (%) among surviving nodes vs Node Failure Rate (P_f).

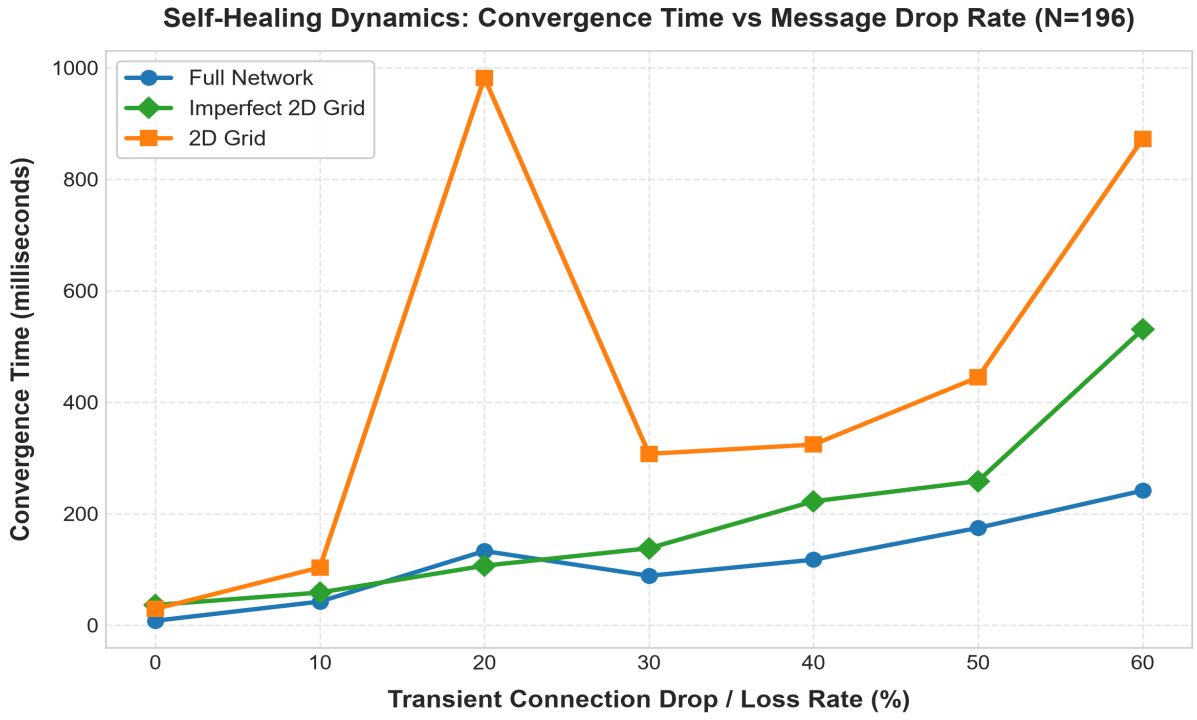


Figure 4: Convergence Time (ms) vs Message Drop Rate (P_d) showing graceful self-healing.

3. Tabulated Experimental Failure Data (N = 196)

Table 3: Rumor Coverage & Convergence Time under Node Crash Failures (N = 196)

Failure Rate (P_f)	Full Network	Imperfect 2D (imp2D)	2D Grid	Line
0% (Healthy)	100% (9.3 ms)	100% (32.1 ms)	100% (41.1 ms)	100% (646.6 ms)
5% Failures	100% (35.1 ms)	100% (37.4 ms)	100% (61.8 ms)	**7.2%** (Partitioned)
10% Failures	100% (47.1 ms)	100% (55.9 ms)	100% (117.6 ms)	**1.1%** (Cut off)
15% Failures	100% (60.7 ms)	100% (112.7 ms)	99.0% (Localized holes)	4.2% (Cut off)
20% Failures	100% (52.6 ms)	100% (138.2 ms)	99.6% (Route detour)	2.3% (Cut off)
25% Failures	100% (86.4 ms)	100% (291.9 ms)	97.3% (Minor cut)	2.3% (Cut off)
30% Failures	100% (108.7 ms)	**99.3%** (Resilient)	**65.9%** (Percolation drop)	1.7% (Cut off)
40% Failures	100% (116.5 ms)	**96.0%** (Robust)	**43.8%** (Disconnected)	0.8% (Cut off)

Table 4: Convergence Time under Transient Connection Loss (N = 196)

Message Drop Rate (P_d)	Full Network Time	Imperfect 2D Time	2D Grid Time	Coverage Achieved
0% (No Loss)	7.99 ms	36.25 ms	29.29 ms	100.0%
10% Loss	42.39 ms	58.64 ms	103.70 ms	100.0%
20% Loss	133.09 ms	106.77 ms	981.88 ms	100.0%
30% Loss	88.68 ms	138.07 ms	307.47 ms	100.0%
40% Loss	117.56 ms	222.28 ms	324.03 ms	100.0%
50% Loss	174.63 ms	258.56 ms	444.79 ms	100.0%
60% Loss	241.43 ms	530.47 ms	872.07 ms	100.0%

4. Key Discoveries & Percolation Dynamics

Observation 1: Zero-Percolation Catastrophe in the Line Topology:
In a 1D line topology, the site percolation threshold is theoretically $p_c = 1.0$. Every node is a single point of failure (cut vertex). Empirically, even a 5% node failure rate cuts coverage down to **7.2%**, and a 10% failure rate reduces coverage to **1.1%**. The rumor immediately halts upon encountering the first failed node in the sequence, isolating the remaining 90%+ of the network.

Observation 2: Site Percolation Phase Transition in 2D Grid:
On a 2D square lattice, the critical site percolation threshold is p_c *approx 0.5927* (i.e. failure threshold ~40.7%). Our data beautifully captures this phase transition: between 0% and 20% failure, the 2D grid routes around dead nodes with **99.6%** coverage. However, as failure rate increases to 30% and 40%, the grid fractures into isolated clusters, and coverage plunges sharply to **65.9%** and **43.8%** respectively.

Observation 3: Shortcut-Driven Fault Immunity of Imperfect 2D:
The Imperfect 2D grid exhibits extraordinary resilience compared to the standard 2D grid. Even at **40% node failure**,

imp2D maintains **96.0% coverage** among living nodes (compared to only 43.8% for standard 2D). The random long-range shortcut edges bridge across partitioned spatial boundaries, preserving giant component connectivity even when the underlying geometric lattice is shattered.

Observation 4: Self-Healing Invariance under Packet Loss:
Under transient link loss, Gossip demonstrates perfect self-healing: **100% of nodes converge across all loss rates up to 60%**. Because active nodes periodically retry transmission to randomly chosen neighbors, dropped packets simply act as stochastic delays. Convergence time scales smoothly as $O(1 / (1 - P_d))$ without requiring costly end-to-end ACKs.